

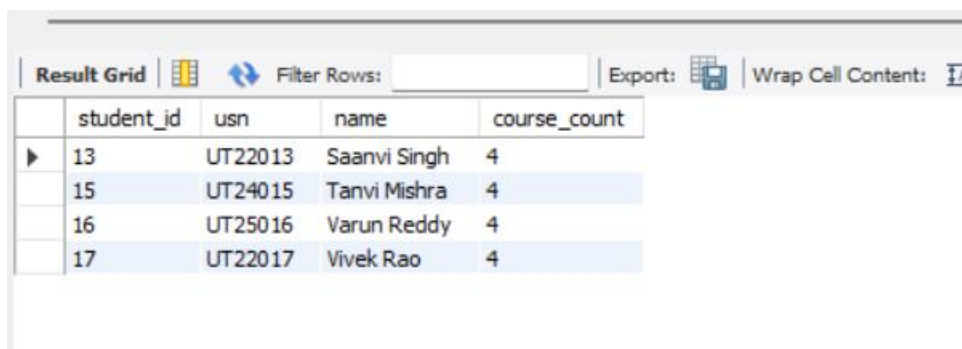
SQL Queries and Results:

This document contains the business questions, SQL query explanations, and results obtained from the UniTrack database.

- Q1. Which students are enrolled in the most courses?

Explanation: This query counts the number of course enrollments for each student. It then finds the highest enrollment count and displays all students who have that highest count.

Output:



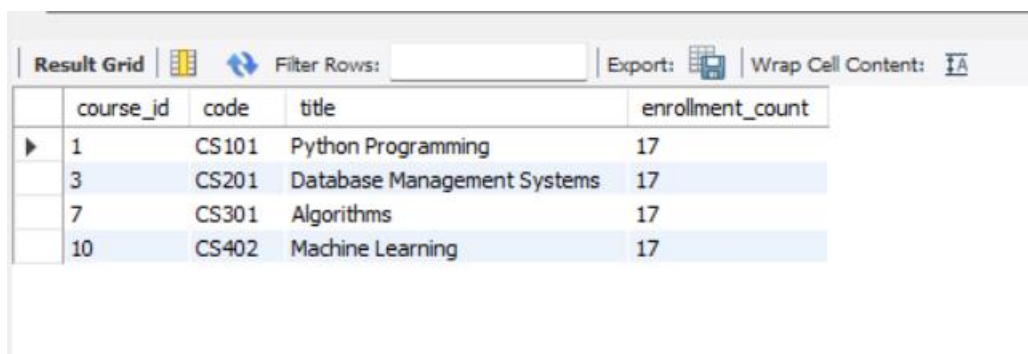
	student_id	usn	name	course_count
▶	13	UT22013	Saanvi Singh	4
	15	UT24015	Tanvi Mishra	4
	16	UT25016	Varun Reddy	4
	17	UT22017	Vivek Rao	4

Result: Four students have the highest number of course enrollments. Each of them is enrolled in 4 courses.

-- Q2. Which courses have the highest enrolment?

Explanation: This query counts the number of students enrolled in each course. It compares the enrollment counts and displays the courses with the highest number of enrollments.

Output:



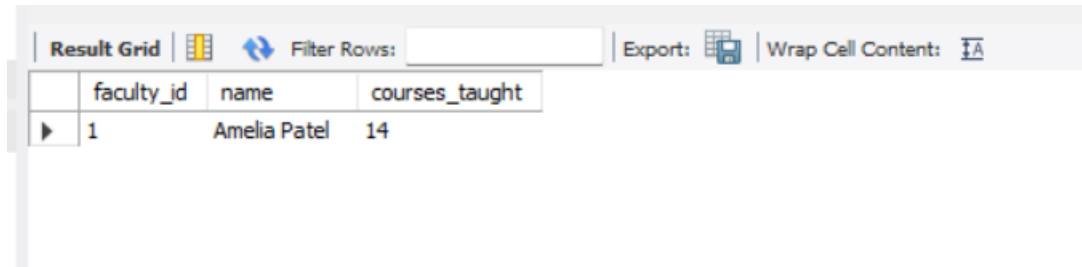
	course_id	code	title	enrollment_count
▶	1	CS101	Python Programming	17
	3	CS201	Database Management Systems	17
	7	CS301	Algorithms	17
	10	CS402	Machine Learning	17

Result: The courses shown in the output have the highest number of student enrollments.

-- Q3. Which faculty members teach the most courses?

Explanation: This query counts the number of course offerings taught by each faculty member. It then finds the highest count and displays all faculty members who teach that number of courses.

Output:



The screenshot shows a database interface with a 'Result Grid' tab. The grid has three columns: 'faculty_id', 'name', and 'courses_taught'. There is one row of data with the values 1, Amelia Patel, and 14. Above the grid, there are controls for 'Filter Rows' (a text box), 'Export' (a button with a document icon), and 'Wrap Cell Content' (a button with a text icon).

faculty_id	name	courses_taught
1	Amelia Patel	14

Result: The faculty member shown in the output teach the highest number of courses in the database.

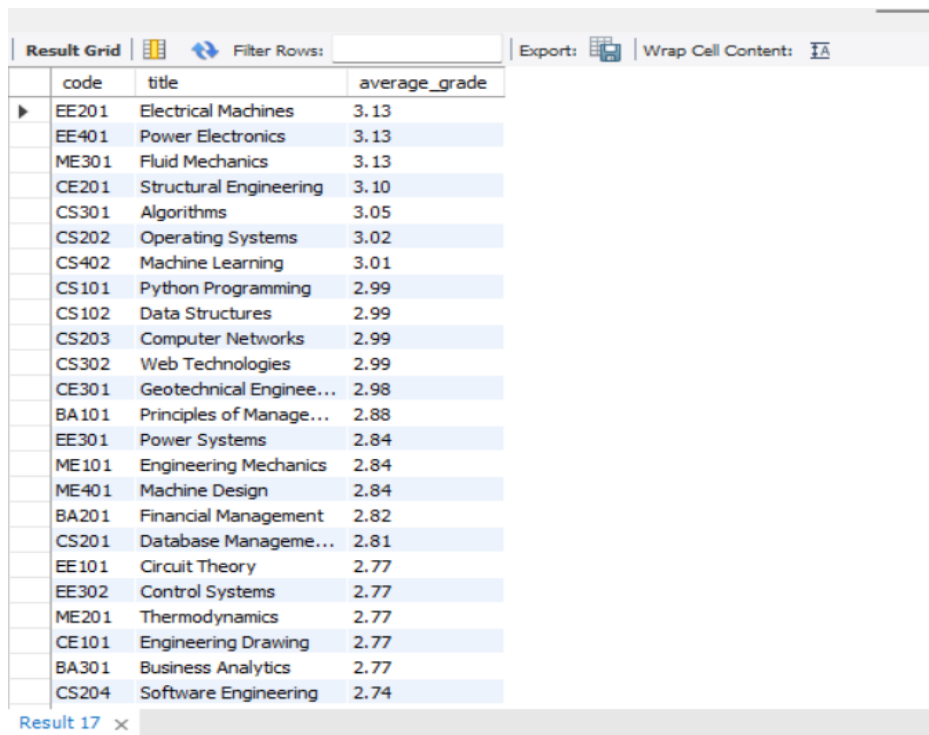
-- Q4. What is the average grade for each course?

Explanation: This query calculates the average grade for each course by converting letter grades into grade-point values. The 'In Progress' grades are excluded because they are not final grades.

The grade-point mapping used is:

- A = 4.0
- A- = 3.7
- B+ = 3.3
- B = 3.0
- B- = 2.7
- C+ = 2.3
- C = 2.0
- D = 1.0
- F = 0.0

Output:



The screenshot shows a 'Result Grid' window with a table of course data. The table has three columns: 'code', 'title', and 'average_grade'. The data is sorted by average grade in descending order. The window includes a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Content' checkbox. The table is titled 'Result 17'.

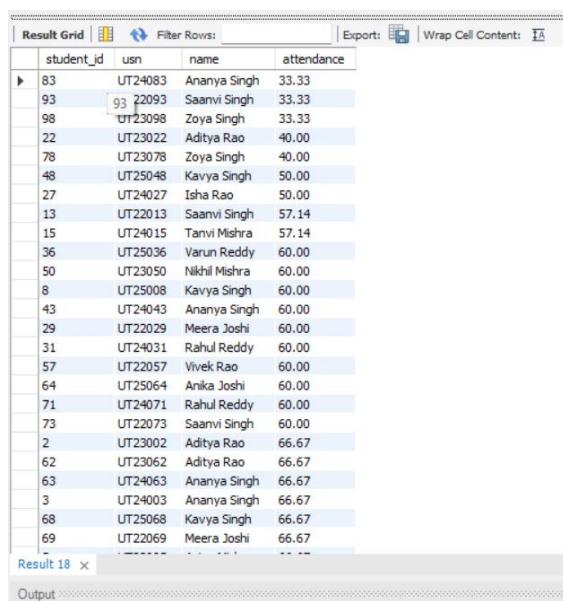
code	title	average_grade
EE201	Electrical Machines	3.13
EE401	Power Electronics	3.13
ME301	Fluid Mechanics	3.13
CE201	Structural Engineering	3.10
CS301	Algorithms	3.05
CS202	Operating Systems	3.02
CS402	Machine Learning	3.01
CS101	Python Programming	2.99
CS102	Data Structures	2.99
CS203	Computer Networks	2.99
CS302	Web Technologies	2.99
CE301	Geotechnical Enginee...	2.98
BA101	Principles of Manage...	2.88
EE301	Power Systems	2.84
ME101	Engineering Mechanics	2.84
ME401	Machine Design	2.84
BA201	Financial Management	2.82
CS201	Database Manageme...	2.81
EE101	Circuit Theory	2.77
EE302	Control Systems	2.77
ME201	Thermodynamics	2.77
CE101	Engineering Drawing	2.77
BA301	Business Analytics	2.77
CS204	Software Engineering	2.74

Result: The output shows the average grade point obtained by students for each course.

-- Q5. Which students have low attendance?

Explanation: This query calculates each student's attendance percentage using the number of 'Present' attendance records divided by the total number of attendance records. Students with an attendance percentage below 75% are displayed.

Output:



The screenshot shows a 'Result Grid' window with a table of student attendance data. The table has four columns: 'student_id', 'usn', 'name', and 'attendance'. The data is sorted by attendance percentage in ascending order. The window includes a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Content' checkbox. The table is titled 'Result 18'.

student_id	usn	name	attendance
83	UT24083	Ananya Singh	33.33
93	22093	Saanvi Singh	33.33
98	UT23098	Zoya Singh	33.33
22	UT23022	Aditya Rao	40.00
78	UT23078	Zoya Singh	40.00
48	UT25048	Kavya Singh	50.00
27	UT24027	Isha Rao	50.00
13	UT22013	Saanvi Singh	57.14
15	UT24015	Tanvi Mishra	57.14
36	UT25036	Varun Reddy	60.00
50	UT23050	Nikhil Mishra	60.00
8	UT25008	Kavya Singh	60.00
43	UT24043	Ananya Singh	60.00
29	UT22029	Meera Joshi	60.00
31	UT24031	Rahul Reddy	60.00
57	UT22057	Vivek Rao	60.00
64	UT25064	Anika Joshi	60.00
71	UT24071	Rahul Reddy	60.00
73	UT22073	Saanvi Singh	60.00
2	UT23002	Aditya Rao	66.67
62	UT23062	Aditya Rao	66.67
63	UT24063	Ananya Singh	66.67
3	UT24003	Ananya Singh	66.67
68	UT25068	Kavya Singh	66.67
69	UT22069	Meera Joshi	66.67

Result: The students shown in the output have attendance below the 75% threshold.

-- Q6. Which students have not submitted an assignment?

Explanation: This query checks the assignments associated with the courses in which each student is enrolled. A `LEFT JOIN` is used to find student-assignment combinations where no submission record exists. If the submission ID is `NULL`, it means that the student has not submitted that assignment.

Output:

student_id	usn	name	assignment_id	title
91	UT24091	Rahul Reddy	42	Assignment 2
92	UT25092	Riya Rao	12	Assignment 4
92	UT25092	Riya Rao	13	Assignment 1
92	UT25092	Riya Rao	27	Assignment 3
92	UT25092	Riya Rao	42	Assignment 2
92	UT25092	Riya Rao	43	Assignment 3
93	UT22093	Saanvi Singh	13	Assignment 1
93	UT22093	Saanvi Singh	27	Assignment 3
93	UT22093	Saanvi Singh	43	Assignment 3
94	UT23094	Siddharth Joshi	13	Assignment 1
94	UT23094	Siddharth Joshi	27	Assignment 3
95	UT24095	Tanvi Mishra	13	Assignment 1
95	UT24095	Tanvi Mishra	43	Assignment 3
97	UT22097	Vivek Rao	13	Assignment 1
97	UT22097	Vivek Rao	43	Assignment 3
98	UT23098	Zoya Singh	28	Assignment 4
98	UT23098	Zoya Singh	43	Assignment 3
99	UT24099	Neha Joshi	13	Assignment 1
99	UT24099	Neha Joshi	14	Assignment 2
99	UT24099	Neha Joshi	28	Assignment 4
99	UT24099	Neha Joshi	43	Assignment 3
99	UT24099	Neha Joshi	44	Assignment 4
100	UT25100	Karan Mishra	14	Assignment 2
100	UT25100	Karan Mishra	28	Assignment 4
100	UT25100	Karan Mishra	44	Assignment 4

Result: The output lists the students and assignments for which no submission record was found.

-- Q7. Which courses currently have no enrolment?

Explanation: This query uses `LEFT JOIN` to include all courses, even when there are no students enrolled in their course offerings. Courses with zero enrollment are displayed in the result.

Output:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
course_id	code	title	

Result: All courses in the database currently have at least one student enrollment. Therefore, there are no courses with zero enrollment.

-- Q8. Which courses have the highest average assignment marks?

Explanation: This query calculates the average marks obtained by students for assignments in each course. It then compares the average marks and displays the courses with the highest average assignment marks.

Output:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
code	title	average_marks	
BA201	Financial Management	24.90	

Result: The course shown in the output have the highest average assignment marks.

-- Q9. Which semester and year has the highest number of enrollments?

Explanation: This query checks how many students are enrolled in courses for each **semester and year**.

It groups the enrollments based on the semester and year of the course offering and counts the number of enrollments in each group.

Finally, it identifies the **semester and year with the highest number of enrollments**.

Output:

Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	year	semester	enrollment_count			
▶	2024	Fall	100			

Result: The semester and year shown in the output has the highest number of student enrollments.

-- Q10. Which courses have the highest number of recorded absences?

Explanation: This query counts the attendance records marked as `Absent` for each course. It then finds the highest absence count and displays all courses having that count.

Output:

Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	code	title	absent_count			
▶	CS201	Database Management Systems	6			
	CS402	Machine Learning	6			

Result: The courses shown in the output have the highest number of recorded student absences.